

CONTACT INFORMATION	✉ E-mail: doserjef@msu.edu Tel: (585) 683-4170 Homepage: https://www.jeffdoser.com GitHub: https://github.com/doserjef
RESEARCH INTERESTS	Statistical Ecology, Wildlife and Natural Resource Conservation, Bayesian Modeling, Biodiversity
EDUCATION	Michigan State University, East Lansing, MI • Ph.D., Forestry and Ecology, Evolution, and Behavior Completed 2022 • Dissertation title: Development and application of hierarchical models for monitoring avian soundscapes, populations, and communities • Advisor: Andrew O. Finley • M.S., Applied Statistics Completed 2021 State University of New York at Geneseo, Geneseo, NY. • B.S. Mathematics and Biology Summa Cum Laude Completed 2018
RESEARCH EXPERIENCE	Postdoctoral Research Associate, Michigan State University 2022 - Current • Department of Integrative Biology • Advisor: Elise F. Zipkin Postdoctoral Fellow: Institute of Biodiversity, Ecology, Evolution, and Macrosystems, Michigan State University 2022 - 2023 Graduate Research Assistant, Michigan State University 2018-2022 • Department of Forestry • Advisor: Andrew O. Finley Undergraduate Research Assistant, SUNY Geneseo 2015-2018 • Advisors: Kristina M. Hannam, David D. Meisel
TEACHING EXPERIENCE	Guest Lecturer, Michigan State University • IBIO 860: Modern Statistical Models in Ecology Feb 2023 • Two guest lectures on capture-recapture models Co-Instructor, Michigan State University • IBIO 831 Statistical Methods in Ecology and Evolution II 2022 • FOR/STT 875 R Programming for Data Sciences 2020-2022 – Lead author of course textbook. Teaching Assistant, Michigan State University • FOR/STT 875 R Programming for Data Sciences 2018-2020 • FOR 472 Ecological Monitoring and Data Analysis 2018 Undergraduate Lab Assistant, SUNY Geneseo 2016 • Introductory Biology Lab

19. **Doser, J.W.**, Finley, A.O., Kéry, M., Zipkin, E.F. **spAbundance**: An R package for single-species and multi-species spatially-explicit abundance models. *Methods in Ecology and Evolution*. Preprint: [arXiv:2310.19446](https://arxiv.org/abs/2310.19446). *In review*.
18. **Doser, J.W.**, Finley, A.O., Saunders, S.P., Kéry, M., Weed, A.W., Zipkin, E.F. Modeling complex species-environment relationships through spatially-varying coefficient occupancy models. *Journal of Agricultural, Biological, and Environmental Statistics*. Preprint: [arXiv:2308.02348](https://arxiv.org/abs/2308.02348). *In review after minor revision*.
17. **Doser, J.W.**, Kéry, M., Saunders, S.P., Finley, A.O., Bateman, B.L., Grand, J., Reault, S., Weed, A.W., Zipkin, E.F. Guidelines for the use of spatially-varying coefficients in species distribution models. *Global Ecology and Biogeography*. Preprint: [arXiv:2301.05645](https://arxiv.org/abs/2301.05645). *In review after major revision*.
16. **Doser, J.W.**[†], Stoudt, S.[†] “Fractional replication” in single-visit multi-season occupancy models: Impacts of spatio-temporal autocorrelation on identifiability. *Methods in Ecology and Evolution*. *In review after major revision*. [†]Authors contributed equally.
15. Gilbert, N.A., Amaral, B.R., Smith, O.M., Williams, P.J., Ceyzyk, S., Ayebare, S., Davis, K.L., Leuenberger, W., **Doser, J.W.**, Zipkin, E.F. A century of statistical Ecology. *Ecology*. *In review*.
14. Bajcz, A.W., Glisson, W.J., **Doser, J.W.**, Larkin, D.J., Fieberg, J.R. A within-lake occupancy model for starry stonewort, a stealthy aquatic invader, to support early detection and monitoring. *Scientific Reports*. *In review*.
13. Kovalenko, V.* , **Doser, J.W.**, Bate, L.J., Six, D.L. Paired acoustic recordings and point count surveys reveal Clark’s Nutcracker and Whitebark Pine associations across a protected area. *Ecology and Evolution*. *In review*. *Graduate student mentee.

12. Zipkin, E.F., **Doser, J.W.**, Davis, C.L., Leuenberger, W.L., Ayebare, S., Davis, K.L. Integrated community models: A framework combining multi-species data sources to estimate the status, trends, and dynamics of biodiversity. *Journal of Animal Ecology*. <https://doi.org/10.1111/1365-2656.14012>.
11. Roberts, C.P., **Doser, J.W.**, Berry, L.L., Fowler, A., Marshall, P.M., Middaugh, C., Rowe, K., Schmit, J., Shaw, M., Wilson, K (2023). Scenario planning and multi-species occupancy models reveal positive avian responses to restoration of afforested woodlands. *Restoration Ecology*. <https://doi.org/10.1111/rec.13998>.
10. Ayebare, S.* , **Doser, J.W.**, Plumptre, A., Owunji, I., Mugabe, H., Zipkin, E.F (2023). An environmental habitat gradient and within-habitat segregation enable co-existence among ecologically similar bird species. *Proceedings of Royal Society B*. 290(2005), 20230467. <https://doi.org/10.1098/rspb.2023.0467>. *Graduate student mentee.
9. **Doser, J.W.**, Finley, A.O., Banerjee, S (2023). Joint species distribution models with imperfect detection for high-dimensional spatial data. *Ecology*. 104(9): e4137. <https://doi.org/10.1002/ecy.4137>.
8. **Doser, J.W.**, Finley, A.O., Kéry, M., Zipkin, E.F (2022). **spOccupancy**: An R package for single-species, multi-species, and integrated spatial occupancy models. *Methods in Ecology and Evolution*. 13(8), 11670-1678. <https://doi.org/10.1111/2041-210X.13897>.
7. **Doser, J.W.**, Leuenberger, W., Sillett, T.S., Hallworth, M.T., Zipkin, E.F (2022). Integrated community occupancy models: A framework to assess occurrence and biodiversity dynamics using multiple data sources. *Methods in Ecology and Evolution*. 13(4), 919-932. <https://doi.org/10.1111/2041-210X.13811>.
6. **Doser, J.W.**, Weed, A.S., Zipkin, E.F., Miller, K.M., Finley, A.O. (2021). Trends in bird abundance differ among protected forests but not bird guilds. *Ecological Applications* 31(6):e02377. <https://esajournals.onlinelibrary.wiley.com/doi/abs/10.1002/eap.2377>.
5. **Doser, J. W.**, Finley, A. O., Weed, A. S., Zipkin, E. F. (2021). Integrating automated acoustic vocalization data and point count surveys for estimation of bird abundance. *Methods in Ecology and Evolution*, 12(6), 1040-1049. <https://doi.org/10.1111/2041-210X.13578>.

4. Hoffmann, H.M., Meadows, J.D., Breuer, J.A., Yaw, A.M., Nguyen, D., Tonsfeldt, K.J., Chin, A.Y., Devries, B.D., Khan, R., Trang, C., Oosterhouse, H.J., Sora Lee, J., **Doser, J.W.**, Gorman, M. R., Welsh, D.K., Mellon, P. L. (2021). The transcription factors VAX1 and SIX3 are required for suprachiasmatic nucleus circadian output and fertility in female mice. *Journal of Neuroscience Research*. <https://doi.org/10.1002/jnr.24864>.
3. Teimouri, M., **Doser, J.W.**, Finley A.O. (2020) ForestFit: An R package for modeling plant size distributions. *Environmental Modelling & Software*, 131, 104668. <https://doi.org/10.1016/j.envsoft.2020.104668>.
2. **Doser, J.W.**, Finley, A.O., Kasten, E.P., Gage, S.H. (2020). Assessing soundscape disturbance through hierarchical models and acoustic indices: A case study on a shelterwood logged northern Michigan forest. *Ecological Indicators*, 113, 106244. <https://doi.org/10.1016/j.ecolind.2020.106244>.
1. **Doser, J.W.**, Hannam, K.M., Finley, A.O., (2020). Characterizing functional relationships between anthropogenic and biological sounds: A western New York state soundscape case study. *Landscape Ecology*, 35(3), 689-707. <https://doi.org/10.1007/s10980-020-00973-2>.

IN PREPARATION

- **Doser, J.W.**, Itter, M.S., Stanke, H., Domke, G.M., Zipkin, E.F., Finley, A.O. Projected increases in vapor pressure deficit forecast substantial forest biomass declines across the western US.
- **Doser, J.W.**, Edwards, M., Jennelle, C.S. Quantifying spatially-varying and nonlinear occupancy trends yields multi-scale insights on anuran distribution change.

BOOKS UNDER CONTRACT

Finley, A.O. and **Doser, J.W.** *Introduction to Forestry Data Analysis with R*. Chapman & Hall CRC. Expected completion date: May 2024.

SOFTWARE

spAbundance. **Doser, J.W.**, Finley, A.O. (Oct 2023 - current). Univariate and Multivariate Spatial Modeling of Species Abundance. <https://cran.r-project.org/web/packages/spAbundance/index.html>.

spOccupancy. **Doser, J.W.**, Finley, A.O. (Nov 2021 - current). Single-species, Multi-species, and Integrated Spatial Occupancy Models. Downloaded 11,677 times as of November 15, 2023. <https://cran.r-project.org/web/packages/spOccupancy/index.html>.

GRANTS AND SCHOLARSHIPS

Funded: USDA Agriculture and Food Research Initiative: *A national-scale early detection and ecosystem service impact assessment tool for invasive plants*. Statistical consultant. 2023-2027. (\$650,000)

Funded: NSF, DEB, Collaborative Research MRA: *Estimating and forecasting nonstationary, multi-scale climate and land-use effects on avian communities*. Postdoctoral Researcher. 2022-2027. (\$1,306,419)

Funded: US National Park Service. *Developing spatially explicit integrated models to evaluate avian distributions and regional bird population dynamics in eastern US National Parks*. Postdoctoral Researcher and Graduate Student Mentor. 2022-2026. (\$228,084)

Not selected: Northeast Climate Adaptation Science Center, USGS. *Improving bird and bat conservation in forest resilience planning: A Northeast climate change adaptation strategy*. Postdoctoral Researcher.

Barry M. Goldwater Scholarship. 2017-2018. (\$7,500)

Dr. Wendell and Barbara Rhodes Research Award. 2016. (\$8,150)

Geneseo Foundation Scholarship. 2014-2018. (\$65,000)

AWARDS AND HONORS	David R. Anderson Outstanding Student Paper Award, Biometrics Working Group, The Wildlife Society	2022
	SUNY Chancellor's Award for Student Excellence	2018
	Excellence in Mathematics, SUNY Geneseo	2018
	Dr. Sam Molnar Scholar Athlete Award, SUNYAC	2018
	James Fulton Award for Academic and Athletic Excellence, SUNYAC	2018
	Phi Beta Kappa Honors Society, SUNY Geneseo	2017
	SUNY Chancellor's Scholar-Athlete Award	2016-2018
	Edgar Fellows Honors Program	2014-2018
UNIVERSITY SERVICE	Ecology, Evolution, and Behavior Program Research Symposium Chair, Michigan State University,	2021-2022
	Ecology, Evolution, and Behavior Program Graduate/Postdoc Awards Committee, Michigan State University,	2020-2022
	Ecology, Evolution, and Behavior Program Research Symposium Committee, Michigan State University	2021
	Treasurer, Graduate Student Organization, Department of Forestry, Michigan State University	2019-2020
	Student Athlete Advisory Committee, SUNY Geneseo	2015-2018
EDITORIAL SERVICE	Peer Reviews: <i>Nature Ecology and Evolution</i> (1), <i>Current Biology</i> (1), <i>Methods in Ecology and Evolution</i> (6), <i>Ecological Monographs</i> (2), <i>Ecological Applications</i> (1), <i>Ecography</i> (4), <i>Ecology</i> (2), <i>Conservation Biology</i> (3), <i>Landscape Ecology</i> (1), <i>Biological Invasions</i> (1), <i>Diversity and Distributions</i> (2), <i>Ecological Indicators</i> (2), <i>Animal Conservation</i> (2), <i>Ecological Solutions and Evidence</i> (1), <i>Ecology and Evolution</i> (2), <i>Ornithological Applications</i> (2), <i>Biometrics</i> (1), <i>Biodiversity and Conservation</i> (1), <i>Journal of Ornithology</i> (1), <i>Insect Conservation and Biodiversity</i> (1), <i>PLOS ONE</i> (1), <i>Ekológia</i> (1)	
RECENT AND UPCOMING WORKSHOPS	Doser, J. W. , Kéry, M. June 2024. Spatial modeling of distribution and abundance in species distribution modeling, population ecology and community ecology. Swiss Ornithological Institute. Sempach, Switzerland.	
	Doser, J. W. , Zipkin, E.F. November 2023. Spatially-explicit occupancy modeling with the spOccupancy R package. The Wildlife Society Annual Conference. Louisville, Kentucky. Open-source materials .	
	Finley, A. O., Doser, J. W. May 2023. Scalable Bayesian models and estimation methods for the analysis of big spatial data and spatio-temporal data. Conference on Applied Statistics and Natural Resources. West Lafayette, Indiana. Open-source materials .	
	Doser, J. W. , Finley, A. O. October 2022. Spatially-explicit occupancy modeling with the spOccupancy R package. Ecological Society of America Statistical Ecology Webinar Series. Remote. Recording and open-source materials .	
	Doser, J. W. July 2022. Bayesian occupancy modeling with the spOccupancy R package. Living Landscapes Lab, University of Arkansas. Remote.	
	Doser, J. W. July 2022. Bayesian occupancy modeling with the spOccupancy R package. Tel Aviv University. Tel Aviv, Israel.	
	Doser, J. W. June 2022. Bayesian occupancy modeling with acoustic data in spOccupancy. Cornell Acoustic Methods Working Group Series. Remote.	

INVITED PRE-
SENTATIONS

- Doser, J. W.**, Finley, A. O., Zipkin, E. F. March 2023. High-dimensional spatial models for predicting forest biomass and bird distributions across the continental US. Levi, Finland. BayesComp 2023. Oral presentation.
- Doser, J. W.** January 2023. Hierarchical models for high-dimensional ecological data across large spatial domains. Bucknell University Department of Mathematics Distinguished Visiting Professor Program. Lewisburg, PA. Seminar.
- Doser, J. W.** November 2022. Using spatially-explicit occupancy models to understand occurrence and biodiversity patterns across macroscales. University of Arkansas Biology Seminar Series. Fayetteville, Arkansas. Seminar.
- Doser, J. W.**, Finley, A.O., Banerjee, S. August 2022. Joint species distribution models with imperfect detection for high-dimensional spatial data. Joint Statistical Meetings. Washington, D. C. Oral presentation.

OTHER PRE-
SENTATIONS

- Doser, J. W.**, Leuenberger, W., Sillett, T.S., Hallworth, M.T., Zipkin, E.F. August 2022. Integrated community occupancy models: A framework to assess occurrence and biodiversity dynamics using multiple data sources. Ecological Society of America. Montréal, Canada. Oral presentation.
- Doser, J. W.**, Leuenberger, W., Sillett, T.S., Hallworth, M.T., Zipkin, E.F. June 2022. Integrated community occupancy models: A framework to assess occurrence and biodiversity dynamics using multiple data sources. International Statistical Ecology Conference. Remote. Oral presentation.
- Doser, J.W.**, Weed, A.S., Zipkin, E.F., Miller, K.M., Finley, A.O. July 2021. Trends in bird abundance differ among protected forests but not bird guilds. American Ornithological Society Meeting. Remote. Oral presentation.
- Doser, J.W.**, Finley, A.O., Kasten, E. P., Gage, S. H. June 2021. Assessing soundscape disturbance through hierarchical models and acoustic indices: A case study on a shelterwood logged northern Michigan forest. International Ecoacoustics Congress. Remote. Oral presentation.
- Doser, J.W.**, Weed, A.S., Zipkin, E.F., Miller, K.M., Finley, A.O. May 2021. Trends in bird abundance differ among protected forests but not bird guilds. Ecology, Evolution, and Behavior Program Symposium. Michigan State University. Remote. ***Awarded 2nd Place for Best Oral Presentation.***
- Doser, J.W.** February 2021. Forest bird monitoring using acoustic recordings and hierarchical models. Hanover Seminar Series. Department of Forestry. Michigan State University. Oral presentation.
- Doser, J.W.**, Finley, A.O., Kasten, E. P., Gage, S. H. February 2020. Assessing soundscape disturbance through hierarchical models and acoustic indices: A case study on a shelterwood logged northern Michigan forest. Michigan State University. Oral presentation.
- Doser, J.W.**, Hannam, K.M., Finley, A.O. September 2019. Using hierarchical Bayesian models to analyze acoustic data. XXVII International Bioacoustics Congress, Brighton, England. Oral presentation.
- Doser, J.W.**, Hannam, K.M., Finley, A.O. May 2019. Characterizing functional relationships between biophony and technophony: a western New York soundscape case study. Northeast Regional Environmental Acoustics Symposium. Providence, RI. Oral presentation.
- Doser, J.W.**, Meisel, D.D. April 2018. Developing an automated bird song recognition system using wavelets and simple machine learning techniques. Geneseo Recognizing Excellence, Achievement and Talent Day, Geneseo, NY. Oral presentation.
- Doser, J.W.**, Hannam, K.M. October 2017. Using soundscape maps to assess a local soundscape. Middle States Division American Association of Geographers Annual Meeting, Geneseo, NY. Poster presentation.
- Doser, J.W.**, Hannam, K.M. April 2017. Analysis of seasonal temporal variation of soundscapes in western New York. Northeast Natural History Conference, Cromwell, CT. Oral presentation.
- Doser, J.W.**, Hannam, K.M. August 2016. Use of soundscape recordings to analyze daily temporal variation in avian vocalizations in western New York. North American Ornithological Conference, Washington D.C. Poster presentation.

PROFESSIONAL POSITIONS	<i>Statistical Consultant</i> , Michigan State University	2019-2020
	• College of Agriculture and Natural Resources Statistical Consulting Center • Provided statistical and programming guidance to undergraduate students, graduate students, and faculty members.	
	<i>Student Database Programmer</i> , SUNY Geneseo	2016-2018
	• Computing and Information Technologies Department • Developed, modified, and tested Banner applications using SQL, Groovy, SQR, and the Argos Enterprise Reporting System	
RELEVANT SKILLS	<i>Programming Languages/Statistical Software:</i> R, C++, C, JAGS, NIMBLE, Stan <i>Software:</i> Latex/Knitr, R Markdown/bookdown, R Shiny	
GRADUATE STUDENTS MENTORED	Bruna Amaral, Michigan State University	2022-current
	Wendy Leuenberger, Michigan State University	2022-current
	Vladimir Kovalenko, University of Montana	2022-2023
	Samuel Ayebare, Michigan State University	2022
OUTREACH	Volunteer, Delta 4H Wildlife Club, East Lansing, MI	Spring 2019
	Guest Speaker, French Road Elementary School, Brighton, NY • Discussed applications of Raspberry Pi computers and how to use them for sound recording.	2018
AFFILIATIONS	The Wildlife Society	2022-current
	American Statistical Association	2019-current
	Ecological Society of America	2021-current
	American Ornithological Society	2021-current
	International Society of Ecoacoustics	2019-2022
REFERENCES	Andrew O. Finley	
	Michigan State University	
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	Aaron S. Weed	
	National Park Service	
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